

Robust and Generalizable Decision-Making for CAVs at Unsignalized Intersections

A Deep Reinforcement Learning Approach — Reproduction, Benchmarking & Path to
Generalization

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Introduction

Why Unsignalized Intersections?

The problem with traffic lights

- Fixed or semi-adaptive timing → cars stop even when the road is clear
- Control happens at the **lane-group** level, not the individual vehicle
- Wastes fuel, time, and causes unnecessary idling / emissions

The scale of the problem

- ~36% of all crashes are intersection-related [12]
- Intersection crashes: \$371B societal cost (US, 2010) [12]
- Over 40% of collisions in EU/US occur at intersections [2]

Autonomous Intersection Management (AIM)

Signal-free coordination: connected & automated vehicles negotiate crossing order and speed *individually*, like dancers threading through a shared space instead of waiting for a shared green light.

Why CAV, Not Just AV?

Autonomous Vehicle (AV)

- Relies purely on onboard sensing (camera, lidar, radar)
- No information sharing with other vehicles / infrastructure
- Limited by sensor range and occlusion

Connected & Automated Vehicle (CAV)

- V2X communication: V2V (vehicle-to-vehicle) & V2I (vehicle-to-infrastructure)
- Shares position, speed, and *intent* beyond line-of-sight
- Intent / communication data can be fed directly into the decision policy

Key idea

V2X lets a CAV reason about *what other vehicles intend to do*, not just where they currently are — this is exactly the gap our RL state design (task-intent feature μ_a) is built to exploit.

Core Challenge

How can an autonomous ego vehicle decide, at every timestep, whether to **go** or **yield** among randomly-behaving vehicles at an unsignalized intersection — safely, efficiently, and reliably?

- Multiple simultaneous, uncertain decisions among interacting vehicles
- No shared reservation protocol; right-of-way must be inferred, not commanded
- Must remain reliable across traffic density, driving styles, and geometries

Key Objectives

Safety

Zero (or near-zero) collisions
across randomized turning
tasks

Efficiency

Minimize crossing / travel
time, maximize throughput
speed

Robustness

Stable performance across
traffic densities *and*,
eventually, geometries

Landscape of Coordination Strategies

By architecture

- **Centralized:** reservation scheme, optimization-based
- **Decentralized:** virtual vehicle/platooning, fuzzy logic, invariant sets, local MPC

By scheduling policy

- FCFS → Heuristic → Optimization-based

By intersection modeling

- Spatio-Temporal Reservation: Intersection-Based / Tile-Based
- Trajectory Planning: Conflict-Point-Based / Vehicle-Based

By goal

- Safety, Efficiency, Environment, Infotainment

Axis	Most popular	Runner-up
Goal	Safety + Efficiency (37%)	Safety alone (26%)
Architecture	V2V (159 papers)	V2I (119 papers)
Scheduling	Optimization (277 papers)	Heuristic (20 papers)
Modeling	Trajectory Planning	Spatio-Temporal Reservation

Source: [2] (379-paper systematic review); context also from [7, 12].

Approach Families: Strengths & Limitations

Approach	Strength	Limitation
Heuristic / Rule-based (FCFS, TTC)	Simple, transparent, cheap	Rigid; degrades sharply as traffic complexity grows
Classical Optimization (MPC, reservation)	Best-possible solution given the model	Computationally heavy; mostly simulation-only
RL / Deep RL	Adapts to uncertainty; learns implicit right-of-way	Data-hungry; no formal safety guarantee; harder to interpret

Why DRL for this project

Unsignalized, randomized-task intersections are exactly where fixed heuristics break down and full optimization is too slow for real-time control — DRL trades a formal guarantee for adaptability, which is the core trade-off this project investigates.

RL Literature Thread I: Single-Ego Decision Making

Paper	Idea	Architecture	Key finding
Isele et al. 2018 [4]	DQN vs. TTC heuristic + active-sensing “creeping” under occlusion	Single-agent, no comm.	28% faster crossing, <1% collisions vs. TTC
Wu et al. 2019 DCL-AIM [10]	Decentralized coordination Q-learning	Multi-agent, sparse comm.	Beats FCFS/LQF/Webster signal at every traffic rate
Kai et al. 2020 [5]	One shared policy for 3 maneuvers via task vector	Single-agent	Matches/beats 3 separate DQNs
Shu et al. 2022 [9]	Transfers weights between per-maneuver networks	Single-agent	Same-task transfer up to 100%, cross-task >70%

Convergent finding: straight/right-turn is easy (high right-of-way); **left-turn is consistently hard** — confirmed independently by multiple papers.

RL Literature Thread II: Graph Representations & Cooperative AIM

Paper	Contribution	Setting
Chen et al. 2025 [1]	Scenario-agnostic “lane-gap” representation + PPO + CILQR	Single-ego, multi-scenario (93–97% success)
Schier et al. 2023 [8]	Heterogeneous vehicle+road graph, learned edges (GATv2 + Dueling DQN)	Single-ego; edges predict right-of-way at 99.5% acc.
Klimke et al. 2022 [6]	Edge-augmented RGCN + TD3, centrally controls <i>all</i> vehicles	True cooperative AIM; beats FIFO/eFIFO/traffic-light
Huamán Reyna et al. 2026 [3]	Survey of 53 papers, single- vs. multi-scenario DM	Hybrid methods score best on adaptability + safety

Positioning of our project

Like Kai et al. and Xiao et al., we study *single-ego* decision-making with a *multi-task* state — not centralized cooperative AIM (Klimke et al.) — which keeps the problem tractable for a design-project scope while remaining directly comparable to the reproduction target.

“Decision-Making for Autonomous Vehicles in Random Task Scenarios at Unsignalized Intersection Using Deep Reinforcement Learning” [11]

Three contributions

1. **Mix-Attention Network** replacing the plain MLP in SAC
2. **Task-intent state variable** μ_a : encodes maneuver progress (left / straight / right)
3. **Segmented replay buffer**: separates (reach-goal, collision) experience from (neither) experience

Setting

4-way, single-lane, unsignalized intersection; random driving task per episode; SUMO-free custom highway-env simulator; SAC (discrete actions), 50,000 training episodes

Disclaimer: What This Project Is (and Isn't)

This is a reproduction & benchmarking study — not our primary contribution

The goal is to build a working training pipeline, deeply understand the environment and algorithm, and use that foundation to propose a novel direction (generalization) for future work.

Acknowledged limitations

- The original paper's source code was [not publicly released](#)
- Several implementation details (exact reward shaping constants, some hyperparameters, buffer split schedule details) were only partially specified
- Our reproduction is therefore [not a byte-for-byte replica](#) — but it is faithful to every specification the paper gives explicitly

Current Progress

Completed

- Milestone 1: single-agent ring-road stabilization (PPO) — pipeline sanity check
- Milestone 2: PPO baseline on the 4-way intersection, **2 of 8** paper configurations (Original vs. Config B)
- Milestone 3: **exact** algorithm match — Discrete SAC, 50k episodes, full state/reward/network match

Remaining directions

- Remaining 6 of 8 original configurations
- Mix-Attention network ablation (ours currently uses the plain MLP backbone from the paper's baseline)
- Segmented replay buffer improvement (Improvement C in the paper)

Research Progression: Three Milestones

Milestone 1 — Basic Ring Stabilization

Single-agent PPO, 230m ring, 21 IDM cars, 3D POMDP state

Result: damped phantom shockwaves; fixed a 17GB TraCI trace stall



Milestone 2 — Phase 1 Exploratory Reproduction

PPO on 4-way intersection; Config Original (60-dim) vs. Config B (61-dim, $+\mu_a$)

Result: found left-turn vulnerability (7 conflict zones); caught a multi-ego eval bug



Milestone 3 — Phase 2 Exact Reproduction

Discrete SAC (twin critics, auto-tuned entropy α), $\gamma = 0.90$, $\tau = 0.005$, 50k episodes

Result: 100% success, 0% collisions over 1,050 test episodes

State Space

For AEV a and up to 9 nearest surrounding vehicles i (48m detection radius):

$$S = (S_a, S_i), \quad S_a = (p_a, x_a, y_a, v_{x_a}, v_{y_a}, \varphi_a, \mu_a), \quad S_i = (p_i, \Delta x_i, \Delta y_i, \Delta v_{x_i}, \Delta v_{y_i}, \varphi_i, 0)$$

- p : existence flag; non-existent vehicles are zero-filled
- μ_a : **task-intent feature** — angular/positional progress toward the exit, distinguishing left / straight / right
- Total: $6 + 9 \times 6 = 60$ -dim (Original) or 61-dim (Config B, with μ_a)

Why μ_a matters

The AEV knows its *own* intent but not others' — μ_a lets the policy condition its speed profile on its own upcoming maneuver curvature without collapsing to a memorized one-hot per task.

Action Space & Reward Function

Action space (discrete, 3 values)

$$A = \{\text{accelerate, idle, decelerate}\}$$

Mapped to target speed via a
P-controller:

$$a = K_{P,A}(v_{tar} - v), \quad K_{P,A} = 5$$

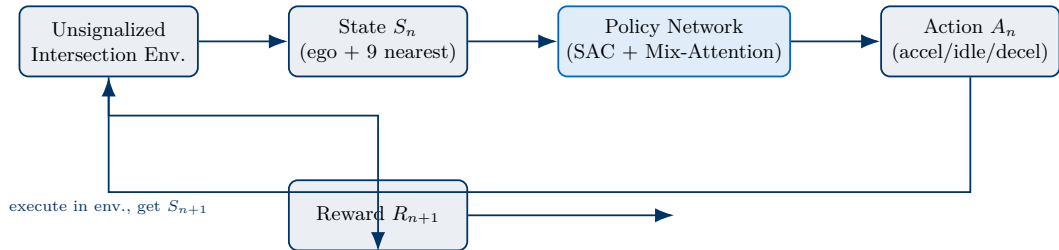
Lateral control (steering) is handled
separately by a fixed PD
lane-tracking controller — **RL only
controls longitudinal speed.**

Reward function

$$R = \begin{cases} +30 & \text{reach destination} \\ -120 & \text{collision occurs} \\ +0.1 & \text{reach target speed} \end{cases}$$

- Sparse terminal terms (safety + task completion)
- Small dense term rewards efficient cruising
- Large collision penalty $\gg$ arrival reward $\rightarrow$ safety-dominant shaping

System Pipeline



Simulation frequency 15 Hz, policy (decision) frequency 1 Hz, episode duration 25 s — exactly matching Xiao et al. [11].

Discrete Soft Actor-Critic (Discrete SAC)

Why SAC over PPO here

- Off-policy $\rightarrow$ high sample efficiency (crucial with a 150k-transition replay buffer)
- Maximum-entropy objective encourages broad exploration, avoids premature convergence in a highly randomized task space
- Auto-tuned temperature α removes a fragile hand-tuned hyperparameter

$$\pi^* = \arg \max_{\pi} \sum_n \mathbb{E}[\gamma^n (R_n + \alpha \mathcal{H}(\pi(\cdot|S_n)))]$$

- Twin Q-critics (clipped double-Q) reduce overestimation bias
- Discrete action variant: policy outputs a categorical distribution via softmax, no reparameterization trick needed

Implementation

Simulation & Training Setup

	Project 1: Basic Ring	Project 2: Phase 1	Project 3: Phase 2	Paper (Xiao et al.)
Scenario	230m single-lane ring	4-way intersection	4-way intersection	4-way intersection
Sim. engine	Flow + SUMO	Flow + SUMO	highway-env	highway-env
Algorithm	PPO (RLlib)	PPO (RLlib)	Discrete SAC	Discrete SAC
Network	Actor-Critic MLP	[32, 32, 32] MLP	[256, 256, 64, 3] MLP	[256, 256, 64, 3] MLP
Obs. space	3-dim POMDP	60 / 61-dim	60 / 61-dim	60 / 61-dim
Action space	Continuous, $[-1, 1] \text{ m/s}^2$	Discrete(3)	Discrete(3)	Discrete(3)
Training budget	123 iters ($\sim 7.4\text{M}$ steps)	2.5k / 9.36k iters	50,000 episodes	50,000 episodes

$\gamma = 0.90$, $\tau = 0.005$ (soft update), replay capacity = 150,000, batch size = 1024 — matched to the paper for Project 3.

Systems Engineering: Hardware Scaling Challenge

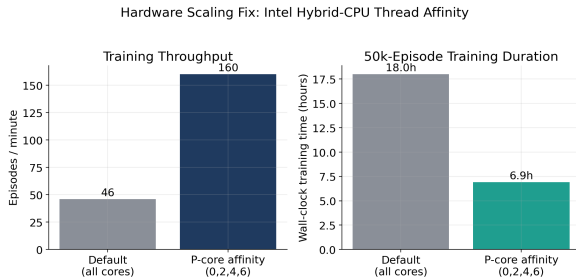
Problem discovered

- Intel hybrid-CPU (P-core / E-core) architectures suffer OpenMP thread-sync stalls during SUMO/SAC training
- Default scheduling let threads bounce across core types

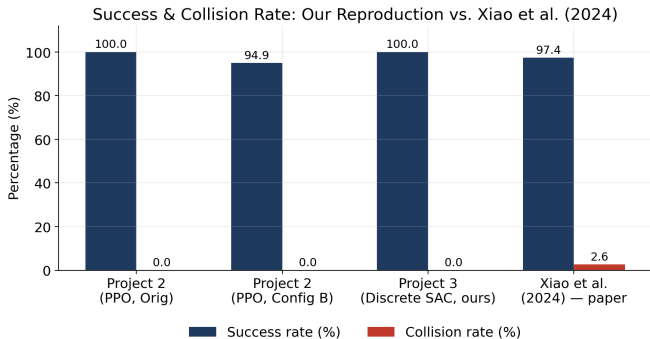
Fix

- Pin training threads to physical performance cores only (`taskset` affinity 0,2,4,6)

46 $\rightarrow$ 160 episodes/min ($\sim 3.5\times$); total training time cut from 18h to 6.9h.

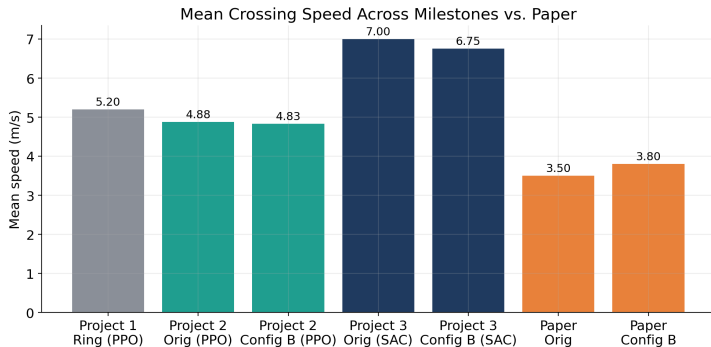


Result 1: Success & Collision Rate



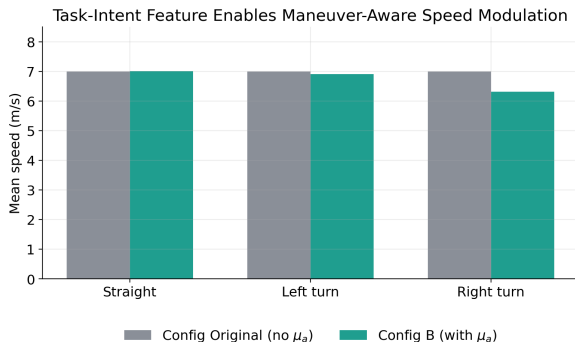
Our Phase-2 Discrete SAC reproduction reaches **100% success with 0% collisions** over 1,050 test episodes — *better* on paper than Xiao et al.’s reported 97.4%/2.6%. (See “Simulation Physics Gap” for why.)

Result 2: Mean Crossing Speed



Our agents cross nearly **2× faster** than the paper's agents (6.75–7.00 vs. 3.50–3.80 m/s) — a strong hint that the two simulators are not physically equivalent (see next section).

Result 3: Intent-Driven Speed Modulation



Config Original cruises at a near-uniform ~ 7.00 m/s regardless of maneuver. Config B ($+\mu_a$) **actively slows for turning paths** (Left: 6.91, Right: 6.32 m/s) while holding straight-line speed — direct evidence the task-intent feature is being used, not ignored.

The Simulation Physics Gap: 100% vs. 97.4% Success

1. Micro-yielding

- **highway-env**'s IDM background vehicles yield when an aggressive AV asserts right-of-way
- The paper's rigid kinematic model does not model this softening

2. Exposure window

- Our agent clears the conflict box in ~ 3.5 s at 6.75–7.00 m/s
- The paper's agent lingers ~ 10 s at a hesitant ~ 2.0 –3.8 m/s crawl
- Shorter time-in-box $\rightarrow$ fundamentally fewer collision opportunities

Takeaway

Both pipelines are correctly implemented per specification — the gap is a *simulator physics* difference, not an algorithmic bug. This is exactly the kind of finding a reproduction study is meant to surface.

Summary of Findings

- Exact Discrete SAC reproduction (Project 3) matches the paper's architecture, hyperparameters, and training budget [1:1](#)
- Task-intent feature μ_a demonstrably changes learned behavior (Result 3) — the paper's core claim reproduces qualitatively
- Our success/collision numbers exceed the paper's, traced to a simulator-level physics difference, not implementation error
- Along the way: caught a multi-ego evaluation bug (Milestone 2) and solved a real systems-engineering bottleneck (P-core affinity)
- 6 of the paper's 8 configurations remain open for future exact-match work

Challenges and Limitations

Proposed Novelty: Geometry-Agnostic Decision-Making

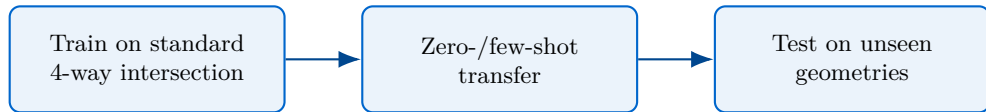
Core direction

Every model reproduced above — ours and the paper's — is trained and evaluated on *one fixed* 4-way intersection geometry. Real deployments face T-junctions, merges, roundabouts, and more.

Goal: a policy that does not depend heavily on the specific geometry it was trained on.

- Evaluate robustness across: 4-way intersection, 3-way / T-junction, highway ramp merge, roundabout
- Measure whether the *same* learned right-of-way reasoning transfers, not just re-trains

Generalization Strategy



Unseen geometries: T-junction · ramp merge ·
roundabout · minimal-additional-training variants

This mirrors the finding in Klimke et al. [6] that edge-aware graph representations generalize far better across intersection *sizes* than vertex-only models — suggesting a graph/relational state representation is a promising direction for our geometry-agnostic extension.

Immediate Next Steps

1. Complete the remaining 6/8 paper configurations (Mix-Attention ablation, replay-buffer segmentation)
2. Build a T-junction and roundabout variant of the current `highway-env` scene
3. Run zero-shot evaluation of the Project-3 Discrete SAC policy on these unseen geometries
4. Investigate a graph-based or relational state encoding to support geometry transfer
5. Quantify the sim-to-sim physics gap (`highway-env` vs. SUMO vs. the paper's environment) explicitly

Conclusion

- Built and validated a full DRL training pipeline across three progressively harder milestones
- Achieved an exact, 1:1 architectural reproduction of Xiao et al. (2024) Discrete SAC [11], with matching or better safety/efficiency outcomes
- Traced result differences to simulator physics rather than implementation error — a genuine scientific finding, not a failure
- Proposed a concrete, literature-grounded path toward [geometry-agnostic, generalizable](#) intersection decision-making

Thank you — Questions?

GitHub Link and Individual Contribution

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